

Robert C. Bridges

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PROFESSIONAL SUMMARY

Aerospace and mechanical engineer with hands-on CAD design experience (Fusion 360) and coursework spanning structural analysis, composite materials, and spacecraft systems. Project work has taken designs from concept through fabrication and test, including a spacecraft systems design carried through SRR, SDR, and PDR review gates. Proficient in Python and MATLAB, with experience coordinating across engineering, quality, and documentation functions on cross-functional teams.

EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Master of Engineering, Aerospace Engineering

May 2026

Bachelor of Science, Mechanical Engineering

Dec 2025

Relevant Coursework: Space Structures; Mechanical Metallurgy; Spacecraft Technology and Systems; Introduction to Spaceflight Mechanics; Propulsion, Aircraft and Rockets; Combustion Processes; Plasmas for Propulsion

TECHNICAL SKILLS

- CAD & Design: Fusion 360, Ansys (FEA/CFD), GD&T
- Materials & Structures: composite materials (honeycomb sandwich panels, carbon fiber layups) and structural analysis fundamentals via coursework in Space Structures and Mechanical Metallurgy
- Programming & Scripting: Python, MATLAB, Java
- Fabrication & Tools: Red Apron Machine Shop, Micro Vu, 3D printing (PLA, TPU, resin), MS Office Suite, Adobe Creative Suite

PROFESSIONAL EXPERIENCE

Abbott Point of Care, Princeton, NJ

Project Engineering Intern

May - Aug 2022 & 2023

- Built and maintained a database of custom parts used across Abbott's medical devices, streamlining search and documentation for engineering teams (2023).
- Analyzed engineering bills of materials and product structures for hundreds of components to support design and manufacturing decisions.
- Used precision measurement equipment, including Micro Vu, to inspect circuit boards; updated and calibrated TEAM, a company-wide database for tracking product errors (2022).
- Documented product issues and test results to strengthen troubleshooting and quality processes throughout the design cycle.

Princeton University, Facilities Department, Princeton, NJ

Engineering Intern

May - Aug 2025 & 2026

- Coordinated communication between project managers and on-site staff, producing daily progress reports to keep stakeholders aligned.
- Read and interpreted technical architectural plans and drafted a granite sign engraving template to support facility construction projects.

ENGINEERING PROJECTS

Original Spacecraft Mission Design, F.O.C.A

Jan - May 2026

- Designed a spacecraft to intercept foreign objects in orbit, using prior flight missions as a baseline for the vehicle architecture.
- Authored SRR, SDR, and PDR documentation to validate the design and defined a full mission life timeline.

Robotic Bat Research, Master's Project

Aug 2025 – May 2026

- Designing and building a robotic bat, including its structure, circuitry, motors, and wing membrane, carrying the hardware from concept through fabrication and test.
- Fabricate and iterate on structural components using PLA, TPU, and resin 3D printing.

Van Allen Belt Electron Simulation

Jan - May 2024

- Modeled the radiation environment that satellites and spacecraft must operate within by simulating electron acceleration in the outer Van Allen Belt during a May 2024 solar storm, recreating observed energies up to 2.4 MeV.
- Simulated electron interaction with whistler-mode chorus waves to explain the observed rise in radiation belt energy.

Focus Glasses

Aug 2025 – Dec 2025

- Led market research and delivered a Product Design Specification with a 4-person team for an innovative solution with the task of keeping focus. Making use of the invisible spectral flicker. Completed finite element analysis to ensure the safety of the glasses.
- Voted best new product in the class after presenting at an exhibit open to the public, professors, and students.

Additional Projects: Spin Wheel, Senior Design, Jan-May 2025 (designed, prototyped, and filed a patent application for a novel computer scroll wheel); Chainsaw dissection, Aug-Dec 2024 (deconstructed chainsaw in order to analyze the fluid dynamic properties).

TEACHING & MENTORSHIP

- Teaching Assistant, Introduction to Spaceflight Mechanics & Introduction to Mechanical Design, Cornell MAE Department: graded assignments, led weekly labs, hosted office hours, and mentored students in aerospace and mechanical design coursework (2025 - Present).
- Grader, Thermodynamics, Cornell MAE Department: created rubrics, graded homework, and delivered performance summaries to faculty (Aug - Dec 2024).

CAMPUS INVOLVEMENT

Ultimate Frisbee, Social Media Chair & Captain (2022-2026) | New Student Orientation Leader (2022) | Mediocre Melodies, Member (2023-2024) | MEng Social Media Content Creator (2026-Present)